Comparative Forecasting of Global Maritime Port Traffic Using Classical and Advanced Time Series Models

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***Abstract*—Accurate forecasting of maritime port traffic is essential for berth planning, labor allocation, congestion mitigation, and supply chain resilience. This study presents a comparative time series forecasting framework for predicting daily vessel calls across major international ports using historical port traffic data from 2019–2026. The selected ports include high-volume and regionally significant maritime hubs such as Singapore, Shanghai, Rotterdam, Busan, Antwerp, Hong Kong, and Japanese industrial ports. Multiple forecasting paradigms are evaluated, including classical statistical models (ARIMA and Holt-Winters), multivariate modeling (VAR), and advanced approaches (Facebook Prophet and XGBoost). Exploratory analysis reveals heterogeneous traffic behavior across ports, including trend shifts, weekly seasonality, disruption-driven anomalies, and weak global cross-port correlations with stronger regional dependencies. Results show that no single model consistently dominates all ports. ARIMA performs well on stable low-volatility series, Prophet excels on strongly seasonal ports, while XGBoost provides superior performance on nonlinear and shock-prone traffic patterns. The findings support adaptive model selection rather than universal forecasting strategies. This work demonstrates how hybrid forecasting pipelines can improve maritime operations and decision-making under uncertain global trade conditions.**

***Index Terms*—Time Series Forecasting, Maritime Analytics, ARIMA, Prophet, XGBoost, Port Traffic Prediction, Supply Chain Intelligence**

I. Introduction

Global seaports are critical nodes in international trade networks. Delays, congestion, and poor planning at ports can create cascading disruptions across logistics systems. Accurate short-term forecasting of vessel arrivals and port traffic enables authorities to optimize berth allocation, workforce scheduling, storage planning, and inland transportation coordination.

Traditional forecasting methods such as ARIMA remain widely used due to interpretability and robustness. However, modern port traffic data often exhibit nonlinear behavior, sudden disruptions, weekly seasonality, and regional interdependencies that challenge purely linear models. This motivates the use of advanced forecasting techniques such as Prophet and machine learning approaches.

The objective of this project is to compare multiple forecasting models on real maritime traffic data and determine which methods are most suitable under different operational conditions.

The major contributions of this study are:

- Comparative evaluation of classical and advanced forecasting models on global port traffic.
- Exploratory identification of trends, anomalies, and inter-port relationships.
- Practical insights into model suitability across stable, seasonal, and volatile ports.
- Real-world interpretation for maritime planning and logistics systems.

II. Literature Review

Time series forecasting has been extensively studied across transportation and logistics domains.

Box and Jenkins introduced ARIMA modeling, which remains a benchmark for linear temporal forecasting. Holt-Winters exponential smoothing is effective for level, trend, and seasonal decomposition in operational planning.

Taylor and Letham proposed Prophet, an additive forecasting model designed for business time series with trend changes and multiple seasonalities. It has been widely adopted due to robustness and ease of tuning.

Chen and Guestrin developed XGBoost, a gradient boosting framework capable of capturing nonlinear feature interactions and lag dependencies. It has shown strong performance in structured forecasting tasks.

Recent studies in maritime analytics have applied machine learning for vessel arrival prediction, congestion detection, and trade volume estimation, indicating that hybrid approaches often outperform single-model pipelines.

However, many existing studies focus on one port or one algorithm. Cross-port comparative forecasting remains relatively underexplored, motivating the present work.

III. Dataset Description and Preprocessing

A. Dataset

The dataset contains daily vessel call observations for multiple globally relevant ports from 2019 to 2026. Representative ports include:

- Singapore

- Shanghai
- Rotterdam
- Antwerp
- Busan
- Hong Kong
- Ningbo
- Japanese industrial ports (Nagoya, Kobe, Yokohama, Chiba, Mizushima)

B. Preprocessing Steps

The following preprocessing pipeline was applied:

- Date parsing and chronological ordering
- Missing value treatment
- Daily frequency alignment
- Port-wise segmentation
- Train-test split preserving temporal order
- Lag feature generation for machine learning models
- Stationarity checks for ARIMA where required

C. Exploratory Findings

Exploratory analysis revealed:

- Singapore as the highest-volume relatively stable port
- Shanghai with higher volatility and episodic declines
- Rotterdam and Antwerp with steady European traffic
- Clear weekly seasonality in several ports
- Event-linked anomalies during COVID-19, Suez blockage, and Red Sea disruptions

IV. Methodology

A. Classical Models

ARIMA: Used for autoregressive linear forecasting after stationarity adjustments.

Holt-Winters: Applied to capture level, trend, and seasonality through exponential smoothing.

B. Multivariate Model

VAR: Used for regionally correlated ports to model joint dependencies among multiple traffic series.

C. Advanced Models

Prophet: Used for automatic trend and seasonal decomposition with changepoint handling.

XGBoost: Trained using lag-based supervised learning features such as:

- Own lag-1 traffic
- Own lag-7 traffic
- Rolling means
- Regional leader lag values
- Calendar effects

D. Evaluation Metrics

Models were evaluated using:

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} \tag{1}$$

$$MAPE = \frac{100}{n}\sum_{i=1}^{n}\left|\frac{y_i - \hat{y}_i}{y_i}\right| \tag{2}$$

RMSE penalizes large deviations, while MAPE enables scale-independent comparison across ports with different traffic volumes.

V. Results and Discussion

A. Comparative Performance

The experiments showed no universally dominant model.

TABLE I
OBSERVED MODEL STRENGTHS

Model	Best Use Case
ARIMA	Stable low-volatility ports
Holt-Winters	Smooth seasonal traffic patterns
Prophet	Strong weekly seasonality and trend shifts
XGBoost	Nonlinear and disruption-prone ports
VAR	Regionally coupled ports

B. Key Findings

- Prophet achieved strong results across several seasonal ports.
- XGBoost performed well on volatile Japanese industrial ports.
- ARIMA remained competitive for smoother traffic series.
- Holt-Winters was reliable but less adaptive to abrupt shocks.
- Weak global correlations suggest most ports should be modeled independently.
- Moderate regional dependencies justify VAR in localized clusters.

C. Interpretation

The findings indicate that port traffic dynamics differ substantially across geography and trade function. Therefore, adaptive model selection is more effective than deploying a single forecasting model across all ports.

VI. Real-World Applications

Accurate port traffic forecasting can support:

- Berth allocation and dock scheduling
- Workforce and crane deployment
- Yard capacity planning
- Congestion mitigation
- Fuel and route optimization
- Early warning during global disruptions

VII. Limitations

This study has several limitations:

- Exogenous drivers such as weather and fuel prices were not included.
- Sudden geopolitical disruptions remain difficult to forecast.
- Deep learning models were not extensively tuned.
- Some ports exhibit highly noisy operational behavior.
- Long-horizon forecasting was outside present scope.

VIII. Conclusion and Future Work

This study compared classical, multivariate, and advanced forecasting models for global maritime port traffic prediction. The results confirm that no single method performs best across all ports. ARIMA remains strong for stable series, Prophet handles seasonality effectively, and XGBoost excels on nonlinear volatile traffic.

Future work may include:

- LSTM/GRU deep learning benchmarks
- Transformer-based temporal models
- Weather and macroeconomic regressors
- Real-time AIS vessel stream integration
- Ensemble forecasting systems

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